SVETLANA VOSKRESENSKAIA

voskr@cosmos.ru **Moscow, Russia**

WORK EXPERIENCE

Space Research Institute (IKI RAS)

Engineer

Nov 2023 – Present

Senior Laboratory Assistant

Jun 2023 – Nov 2023

- Department of High Energy Astrophysics
- Development of machine learning and AI-based algorithms for detecting and investigating galaxy clusters using microwave and X-ray observations

HSE University

Teaching Assistant – Mathematical Analysis

Oct 2020 – Dec 2020

Analytical Approximation Methods

Feb 2021 – Jun 2021

Mechanics

Oct 2022 – Dec 2022

Oct 2023 – Dec 2023

Oct 2024 – Dec 2024

Probability Theory

Oct 2024 – Dec 2024

Astronomy Elective

Feb 2025 – Jun 2025

Feb 2026 – Jun 2026

Gravity and Earth Rotation Elective

Sep 2025 – Dec 2025

HONOURS & AWARDS

- 3rd Prize at the All-Russian Physics Tournament (Russian ITP-2021)
- Increased Faculty Scholarship, Faculty of Physics (2019 – 2023)
- Named Scholarship of the Government of Moscow (2019 – 2023)
- Participant of the 3rd Science Slam, Season IV (2020)
- Informal diploma supplement for participants of the "Cyrillica" project
- Increased Scholarship for Combined Master's-PhD track for HSE students (2023 – Present)

PROFILES

svetlana_voskr
ResearchGate: Svetlana_Voskresenskaia
ORCID: 0009-0000-9338-4047
ResearcherID: JGC-7990-2023
Google Scholar ID: OYlOrP8AAAAJ

ARTICLES

Journal Articles

- Voskresenskaia, S., N. Lyskova, et al. (July 2026). "ComPACT: Mass-redshift properties of the galaxy cluster catalogue". In: 711, A144, A144. DOI: 10.1051/0004-6361/202659201. arXiv: 2605.20027 [astro-ph.CO].
- Voskresenskaia, S., A. Meshcheryakov, and N. Lyskova (June 2024). "ComPACT: combined Atacama Cosmology Telescope + Planck galaxy cluster catalogue". In: 531.1, pp. 1998–2010. DOI: 10.1093/mnras/stae1269. arXiv: 2309.17077 [astro-ph.GA].
- Meshcheryakov, A. V. et al. (Dec. 2023). "SRGz: Classification of eROSITA Point X-ray Sources in the 1{%}DESI Region and Calibration of Photometric Redshifts*". In: *Astronomy Letters* 49.11, pp. 646–661. DOI: 10.1134/S1063773723110129.

Conference Proceedings

- Voskresenskaia, S. A. et al. (2024). "Most likely candidates for galaxy clusters in the PACT catalogue found using deep learning in the submillimetre range". In: *Физика космоса: труды 51-й Всероссийской с международным участием студенческой научной конференции.—Екатеринбург, 2024*. Издательство Уральского университета, p. 110.

EDUCATION

B.Sc. in Physics

HSE University, Moscow

Aug 2019 – Aug 2023

GPA: 9.02
Thesis: Mass Measurements of X-ray Galaxy Clusters Using Data from the Atacama Cosmology Telescope and Deep Learning Approach

Advanced Training in Bioinformatics

HSE University, Moscow

Aug 2020 – Aug 2022

M.Sc. in Physics

HSE University, Moscow

Aug 2023 – Aug 2025

GPA: 8.85 **Thesis**: Neural network catalog of Sunyaev-Zeldovich sources based on ACT+Planck data and analysis of the physical properties of galaxy clusters

CONFERENCES

International Conferences

- Seventeenth Marcel Grossmann Meeting, Talk 8 – 12.7.2024
- The BRICS Astronomy, Talk 8 – 13.9.2024
- High energy astrophysics today and tomorrow, Poster 23 – 26.12.2024
- Russian–Chinese Meeting 2025, Talk 25 – 29.08.2025
- XXIII Conference of Young Scientists "Fundamental and Applied Space Research", Talk 22.4 – 24.4.2026

Russian Conferences

- High energy astrophysics today and tomorrow, Poster 19 – 22.12.2022
- The Youth in Science: Challenges and Prospects, Talk, section winner 14.3.2023
- V Student Conference "From Galactic Nuclei to Atomic Scales", Talk 22 – 25.4.2023
- HSE Computer Science Seminar, Poster 12 – 14.5.2023
- High energy astrophysics today and tomorrow, Talk 18 – 21.12.2023
- All-Russian Student Scientific Conference "Physics of the Cosmos", Talk 29.1 – 2.2.2024
- XXI Conference of Young Scientists "Fundamental and Applied Space Research", Talk 10 – 12.4.2024
- All-Russian Astronomical Conference, Talk 25 – 31.8.2024
- XXII Conference of Young Scientists "Fundamental and Applied Space Research", Talk 21.4 – 25.4.2025
- High energy astrophysics today and tomorrow, Talk 22 – 25.5.2025

RESEARCH PROJECTS

GalaxyHackers – Group Leader

- Development of a deep learning pipeline for detecting galaxy clusters in microwave and infrared surveys

STUDENT SUPERVISION

HSE University – Faculty of Computer Science

- Research Supervisor (Present)
- Supervision of undergraduate students developing neural networks for galaxy cluster detection

School No. 1501

- Research Supervisor (2020 – 2024)
- Led student research projects in astronomy and statistical data analysis

ORGANIZATION OF SCIENTIFIC MEETINGS

High Energy Astrophysics (HEA-2025)

- Conference Organizer (2025)
- Space Research Institute of the RAS (IKI RAS), Moscow, Russia